Anatomy & Embryology: Tongue, Palate, and Oral Anomalies

1. The tongue is a highly mobile muscular organ. Its superior surface (dorsum) is anatomically divided into an anterior two-thirds (oral part) and a posterior one-third (pharyngeal part) by a V-shaped groove called the:
a. Sulcus terminalis
b. Foramen cecum
c. Lingual septum
d. Plica fimbriata
e. Median sulcus
2. At the absolute apex of the V-shaped sulcus terminalis is a small pit that represents an embryonic remnant of the thyroglossal duct origin. This structural landmark is the:
a. Foramen ovale
b. Foramen cecum
c. Incisive fossa
d. Greater palatine foramen
e. Mandibular notch
3. The anterior two-thirds of the tongue dorsum is velvety in texture because it is covered by thousands of mucosal papillae. The most numerous papillae, which are cone-shaped, highly keratinized, and completely lack taste buds, are the:
a. Fungiform papillae
b. Filiform papillae
c. Vallate papillae
d. Foliate papillae
e. Conical papillae
4. The largest papillae on the tongue are arranged in a single V-shaped row immediately anterior to the sulcus terminalis. They contain hundreds of taste buds and are known as the:
a. Fungiform papillae
b. Foliate papillae
c. Filiform papillae
d. Vallate (circumvallate) papillae
e. Conical papillae
5. Histologically, opening into the base of the deep trenches or moats surrounding the circumvallate papillae are the ducts of pure serous minor salivary glands known as:
a. Weber's glands
b. Blandin-Nuhn glands
c. Rivinus glands
d. Glands of Von Ebner
e. Bartholin's glands
6. Fungiform papillae are mushroom-shaped structures scattered primarily at the tip and lateral margins of the tongue. They appear red macroscopically because:
a. They are lined by simple columnar epithelium
b. Their thin, non-keratinized epithelium allows visibility of the underlying vascular capillary loops
c. They contain pure skeletal muscle nests

d. They continuously secrete blood-tinged serous fluid
e. They contain large clusters of melanin granules

7. Foliate papillae appear as parallel vertical mucosal folds or ridges along the posterolateral margins of the tongue. Under the microscope, their lateral walls contain numerous:

a. Taste buds
b. Filiform scales
c. Sebaceous nodes
d. Mucus-secreting goblet units
e. Keratohyalin tufts

8. Microscopic analysis of a taste bud reveals an oval, barrel-shaped intraepithelial receptor organ. The specialized sensory cells that feature apical microvilli projecting into the taste pore are the:

a. Basal (Type IV) stem cells
b. Type II (Light/Receptor) cells
c. Type I supporting cells
d. Langerhans dendritic cells
e. Merkel cells

9. The muscles of the tongue are divided into intrinsic and extrinsic groups. The large, fan-shaped extrinsic muscle that forms the bulk of the tongue substance and acts as the primary protrusor ("safety muscle") of the tongue is the:

a. Hyoglossus
b. Styloglossus
c. Palatoglossus
d. Genioglossus muscle
e. Longitudinalis superior

10. The hyoglossus is a flat, quadrilateral sheet that arises from the greater horn of the hyoid bone. Its primary mechanical action on the tongue is:

a. Protrusion
b. Retraction and elevation
c. Intrinsic narrowing
d. Retraction and depression
e. Lateral twisting

11. The styloglossus muscle arises from the tip of the styloid process and runs forward to insert into the lateral borders of the tongue. Its primary action is to:

a. Protrude the tongue
b. Retract and elevate the tongue upward and backward
c. Depress the floor of the mouth
d. Section the tongue into halves
e. Open the sublingual ducts

12. All intrinsic and extrinsic muscles of the tongue (except palatoglossus) develop from the occipital somites and are supplied motorically by which cranial nerve?

a. Trigeminal nerve (V3 division)
b. Facial nerve (CN VII)
c. Glossopharyngeal nerve (CN IX)
d. Hypoglossal nerve (CN XII)
e. Vagus nerve (CN X)

13. The palatoglossus muscle acts to elevate the base of the tongue and depress the soft palate. Morphologically, it is a muscle of the soft palate and is supplied by the:
a. Mandibular nerve (V3)
b. Facial nerve (CN VII)
c. Glossopharyngeal nerve (CN IX)
d. Vagus nerve (CN X via the pharyngeal plexus)
e. Hypoglossal nerve (CN XII)
14. General sensation (pain, touch, temperature) from the mucosa of the anterior two-thirds of the tongue is carried to the central nervous system via the:
a. Chorda tympani nerve
b. Glossopharyngeal nerve
c. Lingual nerve (branch of V3)
d. Internal laryngeal nerve
e. Hypoglossal nerve
15. Special sensation (taste) from the mucosal papillae of the anterior two-thirds of the tongue is carried by fibers that hitchhike along the lingual nerve but originate from the:
a. Greater petrosal nerve
b. Lesser petrosal nerve
c. Chorda tympani nerve (branch of CN VII)
d. Glossopharyngeal nerve
e. Auriculotemporal nerve
16. Both general sensation and special sensation (taste) from the posterior one-third of the tongue, including the circumvallate papillae, are carried by which cranial nerve?
a. Facial nerve (CN VII)
b. Trigeminal nerve (CN V)
c. Glossopharyngeal nerve (CN IX)
d. Vagus nerve (CN X)
e. Hypoglossal nerve (CN XII)
17. The primary high-pressure arterial blood supply to the tongue tissues is provided by the lingual artery, which branches off the external carotid artery within which cervical zone?
a. Carotid triangle
b. Submental triangle
c. Muscular triangle
d. Digastric triangle (deep to hyoglossus)
e. Occipital triangle
18. Lymphatic drainage of the tongue is highly structured. Lymph fluid from the apex (tip) of the tongue drains bilaterally across the midline to empty directly into the:
a. Submandibular lymph nodes
b. Deep cervical lymph nodes
c. Submental lymph nodes
d. Retropharyngeal lymph nodes
e. Parotid lymph nodes
19. Lymphatic drainage from the lateral margins of the anterior two-thirds of the tongue runs downward to empty into the:
a. Submental lymph nodes

b. Submandibular lymph nodes
c. Mastoid lymph nodes
d. Superficial parotid lymph nodes
e. Pretracheal lymph nodes

20. Lymph from the posterior one-third of the tongue bypasses the superficial nodes and drains directly into the deep cervical chain, primarily targeting the:

a. Submental nodes
b. Jugulo-omohyoid node
c. Jugulodigastric node (tonsillar node)
d. Occipital nodes
e. Submandibular nodes

21. Clinical evaluation of a patient with a lower motor neuron injury of the hypoglossal nerve (CN XII) reveals that when asked to protrude their tongue, the tongue deviates:

a. Toward the paralyzed, injured side
b. Toward the normal, healthy side
c. Alternating side-to-side automatically
d. Straight up toward the hard palate
e. Backward into the pharynx chamber

22. Developmentally, the tongue begins to form around the fourth week of gestation. The anterior two-thirds of the tongue develops from the proliferation of three local mucosal swellings belonging to the:

a. First pharyngeal arch (two lateral lingual swellings and one tuberculum impar)
b. Second pharyngeal arch
c. Third pharyngeal arch
d. Fourth pharyngeal pouch
e. Frontonasal prominence

23. The posterior one-third of the tongue develops from a large midline mucosal elevation situated behind the foramen cecum, known as the copula or hypobranchial eminence. This structure is formed mainly by the mesenchyme of the:

a. First and second pharyngeal arches
b. Third pharyngeal arch (which overgrows the second arch)
c. Fourth and sixth pharyngeal arches
d. Frontonasal process exclusively
e. Laryngeal cartilage templates

24. The transitional zone where the anterior two-thirds of the tongue fuses permanently with the posterior one-third is represented in the adult anatomy by the:

a. Lingual frenulum
b. Sulcus terminalis
c. Median fibrous septum
d. Plica fimbriata
e. Foramen ovale line

25. The hard palate forms the anterior two-thirds of the roof of the mouth. The bony framework of the hard palate is composed of the palatine processes of the maxillae and the horizontal plates of the:

a. Vomer bone
b. Zygomatic bones
c. Sphenoid bone greater wings
d. Palatine bones

e. Ethmoid bone

26. The definitive palate develops from two separate embryonic components. The small, triangular primary palate (median palatine process) is formed directly by the downward growth of the:

a. Maxillary prominence
b. Mandibular process
c. Lateral nasal process
d. Intermaxillary segment (derived from the merged medial nasal processes)
e. Tectal plate

27. The vast majority of the hard palate and the entirety of the soft palate develop from the secondary palate. The secondary palate is formed by a pair of shelf-like mesenchymal projections called the:

a. Medial nasal processes
b. Lateral palatine processes (palatal shelves)
c. Pharyngeal pouches
d. Frontonasal columns
e. Tectal plates

28. The lateral palatine shelves develop from the internal aspects of which embryonic structure?

a. Maxillary prominences
b. First pharyngeal cleft
c. Second pharyngeal arch
d. Mandibular prominence
e. Medial nasal process

29. During the sixth week of development, the lateral palatine shelves project vertically downward on either side of which large midline structural obstruction?

a. Notochord
b. Primary yolk sac
c. Tongue
d. Nasal septum
e. Optic chiasm

30. During the seventh week, the palatal shelves undergo a rapid physiological movement to flip horizontally above the tongue. This critical movement is a:

a. Sliding translation
b. Sagittal contraction
c. Elevation and rotation
d. Retrograde regression
e. Inverted invagination

31. Once the palatal shelves become horizontal, they extend medially to fuse with each other in the midline and with which superior downgrowth?

a. Mandibular symphysis
b. Nasal septum (derived from the frontonasal prominence)
c. Meckel's cartilage
d. Reichert's cartilage
e. Pharyngeal bursa

32. Bone tissue gradually ossifies the anterior segments of the fused palate to create the hard palate, while the posterior segment remains unossified as the muscular:

a. Premaxilla

b. Incisive bone
c. Secondary septum
d. Soft palate and uvula
e. Alveolar process

33. The precise anatomical landmark that marks the boundary between the primary palate and the secondary palate in the adult hard palate is the:
a. Greater palatine foramen
b. Incisive foramen
c. Lesser palatine foramen
d. Pterygoid hamulus
e. Palatine torus

34. The soft palate contains several muscles that regulate its position during swallowing and speech. The muscle that takes origin from the scaphoid fossa and spine of the sphenoid, loops around the pterygoid hamulus, and tenses the palate is the:
a. Levator veli palatini
b. Palatoglossus
c. Palatopharyngeus
d. Tensor veli palatini
e. Musculus uvulae

35. The tensor veli palatini muscle is unique compared to the other palatal muscles because it develops from the first pharyngeal arch and is innervated by the:
a. Facial nerve (CN VII)
b. Mandibular nerve (V3 division of Trigeminal)
c. Glossopharyngeal nerve (CN IX)
d. Vagus nerve (CN X via pharyngeal plexus)
e. Hypoglossal nerve (CN XII)

36. All palatal muscles (except tensor veli palatini) are innervated by motor fibers derived from the cranial accessory nerve traveling through the pharyngeal plexus of the:
a. Oculomotor nerve
b. Vagus nerve (CN X)
c. Trigeminal nerve
d. Facial nerve
e. Glossopharyngeal nerve

37. A complete failure of the lateral palatine processes to fuse with each other in the midline results in a major congenital anomaly known as:
a. Macrocheilia
b. Microstomia
c. Bifid tongue
d. Cleft palate
e. Ankyloglossia

38. If the cleft involves both the primary and secondary palate, extending completely through the alveolar arch to the incisive foramen, it is classified pathologically as a:
a. Median cleft palate
b. Posterior cleft palate
c. Primary palate block
d. Unilateral or bilateral complete cleft palate

e. Submucous cleft palate

39. A mild variation of palatal clefting occurs when the bony and muscular segments fuse normally, but the tissue at the very posterior tip fails to close. This creates a:

a. Cleft lip
b. Macrostomia aperture
c. Bifid uvula
d. Macroglossia defect
e. Thyroglossal cyst

40. When the muscular tissue of the soft palate fails to fuse in the midline but the overlying oral and nasal mucous membranes remain completely intact, the condition is a:

a. Torus palatinus
b. Median cleft lip
c. Submucous cleft palate
d. Branchial fistula
e. Ranula cyst

41. In the midline of the floor of the mouth, the ventral surface of the tongue is anchored to the oral mucosa by a vertical fold called the:

a. Labial frenulum
b. Sublingual fold
c. Lingual frenulum
d. Pterygomandibular raphe
e. Palatine arch

42. An abnormally short or thick lingual frenulum restricts the mobility of the tongue tip, making it difficult to protrude the tongue past the lower incisors. This congenital dental condition is:

a. Macrostomia
b. Ankyloglossia (tongue-tie)
c. Macroglossia
d. Bifid tongue
e. Cleft palate

43. A macroscopic birth defect occurs when the maxillary prominence fails to fuse with the medial nasal prominence on one side. This results in:

a. Median cleft lip
b. Macrostomia
c. Lateral cleft lip (Hare lip)
d. Oblique facial cleft
e. Microstomia

44. An oblique facial cleft is a rare congenital anomaly where the nasolacrimal duct is exposed as an open groove on the surface. This happens due to a total failure of fusion between the:

a. Two mandibular processes
b. Maxillary prominence and lateral nasal prominence
c. Medial and lateral nasal prominences
d. Two medial nasal processes
e. First and second pharyngeal arches

45. A child is born with an abnormally large mouth opening due to a failure of proper developmental fusion between the maxillary and mandibular processes on the lateral aspect. This condition is:
a. Microstomia
b. Macrostomia
c. Mandibulofascial dysostosis
d. Median cleft lip
e. Astomia
46. Severe trauma or a deep fracture passing through the midline symphysis menti can cause a structural tear of the genial tubercles. This causes the genioglossus muscle to lose its anchor, allowing the tongue to:
a. Fall backward into the oropharynx, risking airway obstruction
b. Protrude uncontrollably out of the mouth
c. Deviate purely to the healthy side
d. Undergo immediate calcification
e. Hypertrophy into a macroglossia mass
47. During embryonic life, if the thyroid gland fails to migrate downward from its site of origin at the foramen cecum into the neck, a functional mass of endocrine tissue remains trapped at the base of the tongue, known as a:
a. Thyroglossal duct cyst
b. Lingual thyroid
c. Branchial cyst
d. Ranula
e. Dermoid cyst
48. If a segment of the embryonic thyroglossal tract fails to undergo normal obliteration and continues to secrete fluid postnatally, it presents as a painless, midline fluctuating swelling in the neck that characteristically:
a. Pulsates with the carotid artery
b. Moves upward when the patient protrudes their tongue or swallows
c. Turns completely blue under bright light
d. Radiates severe pain to the mandibular teeth
e. Disappears completely when lying flat
49. Treacher Collins syndrome (mandibulofacial dysostosis) is a severe developmental disorder characterized by malformations of the jaw, zygomatic arch, and ears. This syndrome arises from a failure of normal migration and proliferation of neural crest cells into which specific structures?
a. Second and third pharyngeal pouches
b. Fourth and sixth pharyngeal arches
c. First pharyngeal arch derivatives
d. Frontonasal prominence exclusively
e. Stomodeal ectodermal lining
50. DiGeorge syndrome is a congenital disorder caused by a microdeletion of chromosome 22q11.2. It features immunodeficiency, structural cardiac defects, and hypocalcemia due to a developmental failure of which pouch structures?
a. First and second pharyngeal pouches
b. Third and fourth pharyngeal pouches (leading to absent thymus and parathyroids)
c. First pharyngeal cleft exclusively
d. Frontonasal prominence configurations

e. Ultimobranchial body tracts only

51. Macroglossia refers to a pathological or congenital enlargement of the tongue. It is a classic clinical feature in pediatric patients diagnosed with:

a. Horner's syndrome
b. Bell's palsy
c. Down syndrome (Trisomy 21)
d. Frey's syndrome
e. Early-onset craniosynostosis

52. A rare developmental anomaly of the tongue occurs when the two lateral lingual swellings fail to fuse completely in the midline during the fourth week. This results in a:

a. Lingual thyroid
b. Bifid tongue (glossoschisis)
c. Macroglossia
d. Ankyloglossia
e. Median rhomboid glossitis

53. A localized, smooth, reddish diamond-shaped patch seen in the midline dorsum of the tongue immediately anterior to the sulcus terminalis is median rhomboid glossitis. Clinically, it represents a congenital persistence of the:

a. Tuberculum impar
b. Copula
c. Hypobranchial eminence
d. Lateral lingual swellings
e. First pharyngeal cleft

54. Microscopic examination of a healthy tongue reveals that its bulk is composed of bundles of skeletal muscle fibers. These fibers are arranged in how many planes, allowing for the tongue's high mobility?

a. One single horizontal plane
b. Two perpendicular planes
c. Three interlocking planes (vertical, transverse, and longitudinal)
d. Four diagonal quadrants
e. A random chaotic matrix lacking orientation

55. When a clinician wants to perform an infiltration block of the greater palatine nerve, the needle is directed toward the greater palatine foramen, which is located in the hard palate mucosa medially to which tooth?

a. Maxillary canine
b. Maxillary first premolar
c. Maxillary second molar
d. Maxillary third molar (or between the second and third molars)
e. Maxillary central incisor

56. Successful anesthesia of the greater palatine nerve results in a localized loss of sensation affecting which hard structures?

a. The entire soft palate and uvula
b. The skin of the upper lip and nose
c. The mucosa and palatal gingiva of the hard palate anteriorly up to the canine tooth
d. The anterior two-thirds of the tongue dorsum
e. The lower teeth and mandibular gingiva

57. Sensory block of the anterior-most segment of the hard palate mucosa (behind the incisor teeth) requires a block targeting the nasopalatine nerve at the:
a. Greater palatine foramen
b. Lesser palatine foramen
c. Incisive foramen
d. Mandibular foramen
e. Foramen rotundum
58. A condition characterized by a benign, bony outgrowth or exostosis located in the midline of the hard palate is:
a. Torus palatinus
b. Torus mandibularis
c. Cleft palate
d. Bifid uvula
e. Palatine mucocele
59. Severe, stabbing pain radiating across the palatal and maxillary structures triggered by minor touch or chewing is classic for:
a. Bell's palsy
b. Trigeminal neuralgia (Tic douloureux affecting V2)
c. Glossopharyngeal neuralgia
d. Frey's syndrome
e. Horner's syndrome
60. The minor salivary glands located predominantly within the soft tissue matrix of the soft palate and uvula are histologically classified as:
a. Pure serous glands
b. Pure mucous glands
c. Mixed seromucous glands with abundant demilunes
d. Endocrine columns
e. Sebaceous nests
61. The soft palate elevates and seals off the nasopharynx from the oropharynx during swallowing to prevent food from entering the nasal cavity. This mechanism is called:
a. Velopharyngeal closure
b. Vestibular compression
c. Masticatory cycles
d. Lingual protrusion
e. Tonsillar filtration
62. Microscopic evaluation of the non-keratinized stratified squamous epithelium of the soft palate reveals wide intercellular spaces and a thin lamina propria containing an abundance of:
a. Coarse, rigid bone trabeculae
b. Elastic fibers and minor mucous glands
c. Highly keratinized scales
d. Odontoblast cell bodies
e. Enamel rods
63. The deep venous drainage of the tongue is mediated primarily by the deep lingual vein. This vessel is easily visible clinically on the ventral surface of the tongue, running lateral to the:
a. Sulcus terminalis

b. Lingual frenulum
c. Sublingual fold
d. Circumvallate papillae
e. Palatine arch

64. A complete congenital absence of the palate, which results in a massive, open communication between the oral and nasal cavities from birth, is known as:

a. Microstomia
b. Uranoschisis (complete palatoschisis)
c. Macroglossia
d. Ankyloglossia
e. Astomia

65. When testing the gag reflex, touching the wall of the oropharynx or soft palate triggers a protective contraction. The afferent limb of this reflex is mediated by the glossopharyngeal nerve, and the efferent limb is carried by the:

a. Trigeminal nerve (V3)
b. Facial nerve (CN VII)
c. Vagus nerve (CN X via the pharyngeal plexus)
d. Hypoglossal nerve (CN XII)
e. Accessory nerve (CN XI)

Answer Key Table (Tongue, Palate, Development, and Anomalies -

Q 1-	**Q 9-**	**Q 17-**	**Q 25-**	**Q 33-**	**Q 41-**	**Q 49-**	**Q 57-**

8	16	24	32	40	48	56	65
a	a	c	e	c	c	c	c
b	d	c	b	b	b	b	a
b	b	b	a	e	b	c	b
d	d	c	b	b	c	b	b
d	c	b	d	c	b	d	a
b	e	c	d	c	a	c	b
a	e	a	b	d	b	c	b
b	d	a	b	d	b	c	b
							c